

SAKSHEE UJJWAL KUMAR

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EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science and Engineering Core

2023 – Present

CGPA: 9.03

Srimathi Sundaravalli Memorial School, Chennai

10th Board [CBSE] – 98.4% 12th Board [CBSE] – 95.6%

EXPERIENCE

R&D Intern at PureID

Network Security, Passwordless Authentication, IAM

May 2026 – Present

- Performed network traffic analysis (**Wireshark**) and vulnerability reconnaissance (**Nmap**), correlating fingerprints with **CVEs**; studied **OWASP Top 10** risks and **FIDO2/WebAuthn** passkey authentication.

LLM-Powered SSH Honey pots in AWS

Cyber Defenders Internship

June 2026 – Present

- Deployed and managed a **multi-region system on AWS (EC2, IAM, S3)** across **Singapore and Mumbai**, monitoring live system telemetry and diagnosing behavioral differences between static and AI-assisted configurations.
- Containerized and integrated an **Ollama LLM** backend with the **Beelzebub** framework via **Docker**; engineered automated **S3** log shipping (5-minute sync) for continuous system health visibility across regions.

PROJECTS

CYCLON GOSSIP PROTOCOL

C, UDP Sockets, Distributed Systems, Networking

July 2024 – November 2024

- Implemented and empirically compared two **peer-to-peer gossip protocols** (naive and **Cyclon**) in **C** using **UDP sockets** across **6 concurrent distributed processes**.
- Designed core data structures such as bounded-view descriptor management, **Fisher-Yates**-based random peer sampling, and FIFO message deduplication cache to manage **O(1) view updates** per gossip cycle.
- Derived and validated **closed-form redundancy models** via **linear regression** ($R \approx N$ for fixed fanout, $R \approx 0.73 \cdot N \log_2 N$ for logarithmic fanout).

DRIFTXPLAIN: ML-BASED INTRUSION DETECTION SYSTEM

Python, Scikit-learn, SHAP, Streamlit – CIC-IDS-2017 (2.8M records)

February 2026 – April 2026

- Built a complete IDS pipeline on **2.8M network flow records**, handling severe class imbalance (80.3% benign vs. 19.7% attack, down to 11 samples for rare attacks) via **SMOTE**, class weighting, and threshold tuning.
- Diagnosed model reliability under **concept drift** by temporally splitting data (train: Mon–Wed, test: Thu–Fri) to simulate real deployment – found the highest-test-accuracy model (XGBoost, 99%) **failed completely under drift (0% recall)**, while Random Forest degraded more gracefully, informing model selection based on real-world reliability over benchmark accuracy.
- Deployed the final model (**99.4% recall, 87.7% precision**) as an interactive **Streamlit** app with real-time **SHAP**-based explainability, surfacing which network features drove each classification for root-cause analysis.

NEURODECEPT: ADAPTIVE CYBER DECEPTION

Python, Scikit-learn – Unsupervised Anomaly Detection

July 2025 – October 2025

- Conceived and implemented **NeuroDecept**, an adaptive cyber deception framework integrating unsupervised **Autoencoder anomaly detection** with event-triggered **dynamic graph topology rewiring** in a closed feedback loop, trained on real CICIDS2017 network traffic.
- Achieved **10.75× improvement** in honeypot interception rate over static baselines (43.87% vs 4.08%), with **41.5% security risk reduction** and **2.81× increase in attacker dwell time**, validated across 360 independent simulation trials.
- Conducted rigorous **ablation study** across 4 system configurations, 3 network topologies, and 3 attacker models; statistical significance confirmed at $p = 0.000002$ (Cohen's $d = 2.13$); submitted to **IC CIS2026 (Springer LNNS, Scopus-indexed)**.

TECHNICAL SKILLS

Operating Systems & Networking: Linux, TCP/IP, DNS, HTTP/HTTPS, UDP Sockets, Wireshark

Cloud & Virtualization: AWS, Docker, IBM Cloud

Programming Languages: C, C++, Python, Java, Bash

Core: Agile, DSA, OOP, System Design, Distributed Systems, OS, Computer Architecture, DBMS, Cloud Architecture Design

Databases: SQLite, PostgreSQL, Firebase

LEADERSHIP

Chairperson at VITeach

Led 4 education outreach programs across government schools in partnership with NGOs and coordinated work among various departments.

2024 – Present